THE STANDARD SIMPLICIAL OBJECT OF AN ADJOINT PAIR IS SIMPLICIAL WITH AN AUGMENTATION

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-standard-simplicial (source lines 6891-6898); the proof environment begins two lines below it and is its own proof, closing at source line 6937.

Editorial note. The statement refers to the author's standing situation for the adjoint pair by label; that situation's text is retained as a context span and the generator expands the context transitively, so the reference resolves inside the assembled package.

Editorial note. This card lies in the half of the chapter assigned to this lane under the line-number split (statement start >= 3601); the candidate was selected with that filter before reading.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Objects chapter, counted once: an adjoint pair gives a standard simplicial object of the functor category, augmented at the identity.

Allowed standard prerequisites (editorial).

- Adjoint pair with unit and counit. An adjunction between functors U and V with unit eta and counit epsilon, fulfilling the triangle identities (the author's Situation situation-adjoint-functors, retained as context).. The system X is built from alternating composites of eta and epsilon, and the computations use the triangle identities (source lines 6893-6897, 6900-6937).
- Simplicial object in a functor category. A sequence of functors with face and degeneracy maps satisfying the simplicial identities.. It is what the lemma asserts of the system X and what the proof verifies (source lines 6894-6895, 6905-6933).
- Augmentation to the constant simplicial object. A morphism from a simplicial object to the constant simplicial object at a given value, compatible with all the structure maps.. It is the second assertion of the lemma (source lines 6896-6897).

Source reference. Original lemma 003D: sources/stacks-chow/tags.txt, source lines 130–130. Cross-chapter pointer whose target carries the canonical tag 003D in the

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project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Source reference. Original lemma 003F: sources/stacks-chow/tags.txt, source lines 132–132. Cross-chapter pointer whose target carries the canonical tag 003F in the project tag list; declared so the reference resolves inside the assembled package, while every step the counted argument uses remains inside the retained spans.

Situation 0.1. Let $\mathcal{A}$, $\mathcal{S}$ be categories and let $V : \mathcal{A} \to \mathcal{S}$ be a functor with a left adjoint $U : \mathcal{S} \to \mathcal{A}$.

$$V \xrightarrow{\eta \star 1_V} V \circ U \circ V \xrightarrow{1_V \star d} V \quad \text{and} \quad U \xrightarrow{1_U \star \eta} U \circ V \circ U \xrightarrow{d \star 1_U} U \tag{0.1.1}$$

Example 0.2. Let $Y : \mathcal{C} \to \mathcal{C}$ be a functor from a category to itself and suppose given transformations of functors

$$d : Y \longrightarrow \mathrm{id}_{\mathcal{C}} \quad \text{and} \quad s : Y \longrightarrow Y \circ Y$$

Using these transformations we can construct something that looks like a simplicial object. Namely, for $n \geq 0$ we define

$$X_n = Y \circ \ldots \circ Y \quad (n+1 \text{ compositions})$$

Observe that $X_{n+m+1} = X_n \circ X_m$ for $n, m \geq 0$. Next, for $n \geq 0$ and $0 \leq j \leq n$ we define using notation as in Categories, Section 003D

$$d_j^n = 1_{X_{j-1}} \star d \star 1_{X_{n-j-1}} : X_n \to X_{n-1} \quad \text{and} \quad s_j^n = 1_{X_{j-1}} \star s \star 1_{X_{n-j-1}} : X_n \to X_{n+1}$$

So d_j^n, resp. s_j^n is the natural transformation using d, resp. s on the jth Y (counted from the left) in the composition defining X_n.

Lemma 0.3. *In Example 0.2 if*

$$1_Y = (d \star 1_Y) \circ s = (1_Y \star d) \circ s \quad \textit{and} \quad (s \star 1) \circ s = (1 \star s) \circ s$$

then $X = (X_n, d_j^n, s_j^n)$ is a simplicial object in the category of endofunctors of $\mathcal{C}$ and $d : X_0 = Y \to id_{\mathcal{C}}$ defines an augmentation.

Proof. To see that we obtain a simplicial object we have to check that the relations (1)(a) – (e) of Lemma 0.4 are satisfied. We will use the short hand notation

$$1_a = 1_{X_{a-1}} = 1_Y \star \ldots \star 1_Y \quad (a \text{ factors})$$

for $a \geq 0$. With this notation we have

$$d_j^n = 1_j \star d \star 1_{n-j} \quad \text{and} \quad s_j^n = 1_j \star s \star 1_{n-j}$$

We are repeatedly going to use the rule that for transformations of funtors a, a', b, b' we have $(a' \circ a) \star (b' \circ b) = (a' \star b') \circ (a \star b)$ provided that the $\star$ and $\circ$ compositions in this formula make sense, see Categories, Lemma 003F.

Condition (1)(a) always holds (no conditions needed on d and s). Namely, let $0 \leq i < j \leq n+1$. We have to show that $d_i^n \circ d_j^{n+1} = d_{j-1}^n \circ d_i^{n+1}$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star d \star 1_{n+1-j}) = (1_{j-1} \star d \star 1_{n+1-j}) \circ (1_i \star d \star 1_{n+1-i})$$

We can rewrite the left hand side as

$$\begin{aligned}&(1_i \star d \star 1_{j-i-1} \star 1_{n+1-j}) \circ (1_i \star 1_1 \star 1_{j-i-1} \star d \star 1_{n+1-j})\\ &= 1_i \star ((d \star 1_{j-i-1}) \circ (1_1 \star 1_{j-i-1} \star d)) \star 1_{n+1-j}\\ &= 1_i \star d \star 1_{j-i-1} \star d \star 1_{n+1-j}\end{aligned}$$

The second equality is true because $d \circ 1_1 = d$ and $1_{j-i} \circ (1_{j-i-1} \star d) = 1_{j-i-1} \star d$. A similar computation gives the same result for the right hand side.

We check condition (1)(b). Let $0 \leq i < j \leq n-1$. We have to show that $d_i^n \circ s_j^{n-1} = s_{j-1}^{n-2} \circ d_i^{n-1}$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j-1} \star s \star 1_{n-1-j}) \circ (1_i \star d \star 1_{n-1-i})$$

By the same kind of calculus as in case (1)(a) both sides simplify to $1_i \star d \star 1_{j-i-1} \star s \star 1_{n-j-1}$.

We check condition (1)(c). Let $0 \leq j \leq n-1$. We have to show $\mathrm{id} = d_j^n \circ s_j^{n-1} = d_{j+1}^n \circ s_j^{n-1}$, i.e.,

$$1_n = (1_j \star d \star 1_{n-j}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j+1} \star d \star 1_{n-j-1}) \circ (1_j \star s \star 1_{n-1-j})$$

This is easily seen to be implied by the first assumption of the lemma.

We check condition (1)(d). Let $0 < j+1 < i \leq n$. We have to show $d_i^n \circ s_j^{n-1} = s_j^{n-2} \circ d_{i-1}^{n-1}$, i.e.,

$$(1_i \star d \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_j \star s \star 1_{n-2-j}) \circ (1_{i-1} \star d \star 1_{n-i})$$

By the same kind of calculus as in case (1)(a) both sides simplify to $1_j \star s \star 1_{i-j-2} \star d \star 1_{n-i}$.

We check condition (1)(e). Let $0 \leq i \leq j \leq n-1$. We have to show that $s_i^n \circ s_j^{n-1} = s_{j+1}^n \circ s_i^{n-1}$, i.e.,

$$(1_i \star s \star 1_{n-i}) \circ (1_j \star s \star 1_{n-1-j}) = (1_{j+1} \star s \star 1_{n-1-j}) \circ (1_i \star s \star 1_{n-1-i})$$

By the same kind of calculus as in case (1)(a) this reduces to

$$(s \star 1_{j-i+1}) \circ (1_{j-i} \star s) = (1_{j-i+1} \star s) \circ (s \star 1_{j-i})$$

If $j = i$ this is exactly one of the two assumptions of the lemma. For $j > i$ left and right hand side both reduce to the equality $s \star 1_{j-i-1} \star s$ by calculations similar to those we did in case (1)(a).

Finally, in order to show that d defines an augmentation we have to show that $d \circ (1_1 \star d) = d \circ (d \star 1_1)$ which is true because both sides are equal to $d \star d$. □

Lemma 0.4. *Let $\mathcal{C}$ be a category.*

(1) *Given a simplicial object U in $\mathcal{C}$ we obtain a sequence of objects $U_n = U([n])$ endowed with the morphisms $d_j^n = U(\delta_j^n) : U_n \to U_{n-1}$ and $s_j^n = U(\sigma_j^n) : U_n \to U_{n+1}$. These morphisms satisfy the opposites of the relations displayed in Lemma 0.6, namely*
   (a) *If $0 \leq i < j \leq n+1$, then $d_i^n \circ d_j^{n+1} = d_{j-1}^n \circ d_i^{n+1}$.*
   (b) *If $0 \leq i < j \leq n-1$, then $d_i^n \circ s_j^{n-1} = s_{j-1}^{n-2} \circ d_i^{n-1}$.*
   (c) *If $0 \leq j \leq n-1$, then $\mathrm{id} = d_j^n \circ s_j^{n-1} = d_{j+1}^n \circ s_j^{n-1}$.*
   (d) *If $0 < j+1 < i \leq n$, then $d_i^n \circ s_j^{n-1} = s_j^{n-2} \circ d_{i-1}^{n-1}$.*
   (e) *If $0 \leq i \leq j \leq n-1$, then $s_i^n \circ s_j^{n-1} = s_{j+1}^n \circ s_i^{n-1}$.*

(2) *Conversely, given a sequence of objects U_n and morphisms d^n_j, s^n_j satisfying (1)(a) – (e) there exists a unique simplicial object U in $\mathcal{C}$ such that $U_n = U([n])$, $d^n_j = U(\delta^n_j)$, and $s^n_j = U(\sigma^n_j)$.*
(3) *A morphism between simplicial objects U and U' is given by a family of morphisms $U_n \to U'_n$ commuting with the morphisms d^n_j and s^n_j.*

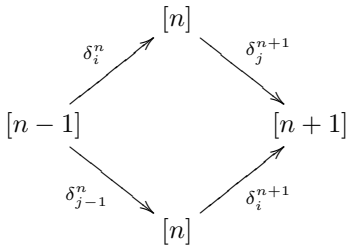
Proof. This follows from Lemma 0.5. □

Lemma 0.5. *The category Δ is the universal category with objects $[n]$, $n \geq 0$ and morphisms δ^n_j and σ^n_j such that (a) every morphism is a composition of these morphisms, (b) the relations listed in Lemma 0.6 are satisfied, and (c) any relation among the morphisms is a consequence of those relations.*

Proof. Omitted. □

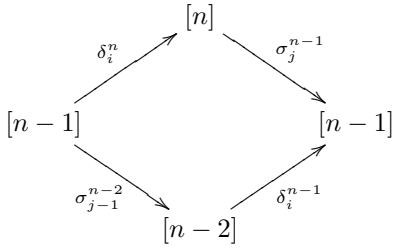
Lemma 0.6. *The morphisms δ^n_j and σ^n_j satisfy the following relations.*

(1) *If $0 \leq i < j \leq n+1$, then $\delta^{n+1}_j \circ \delta^n_i = \delta^{n+1}_i \circ \delta^n_{j-1}$. In other words the diagram*



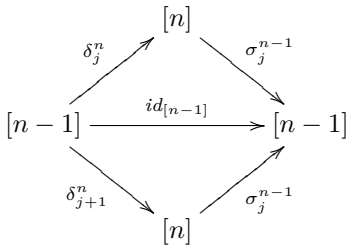
commutes.

(2) *If $0 \leq i < j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_i = \delta^{n-1}_i \circ \sigma^{n-2}_{j-1}$. In other words the diagram*



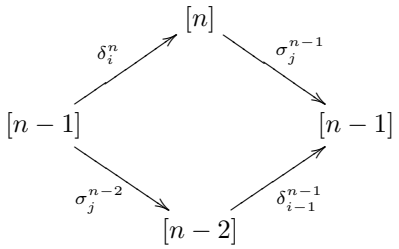
commutes.

(3) *If $0 \leq j \leq n-1$, then $\sigma^{n-1}_j \circ \delta^n_j = id_{[n-1]}$ and $\sigma^{n-1}_j \circ \delta^n_{j+1} = id_{[n-1]}$. In other words the diagram*



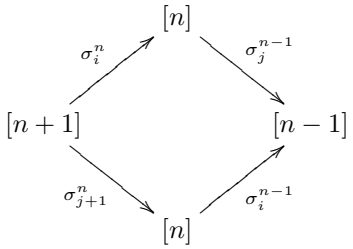
commutes.

(4) *If $0 < j+1 < i \leq n$, then $\sigma_j^{n-1} \circ \delta_i^n = \delta_{i-1}^{n-1} \circ \sigma_j^{n-2}$. In other words the diagram*



commutes.

(5) *If $0 \leq i \leq j \leq n-1$, then $\sigma_j^{n-1} \circ \sigma_i^n = \sigma_i^{n-1} \circ \sigma_{j+1}^n$. In other words the diagram*



commutes.

Proof. Omitted. □

Lemma 0.7. *In Situation 0.1 the system $X = (X_n, d_j^n, s_j^n)$ is a simplicial object of Fun$(\mathcal{A}, \mathcal{A})$ and ϵ_0 defines an augmentation ϵ from X to the constant simplicial object with value $X_{-1} = id_{\mathcal{A}}$.*

Proof. Consider $Y = U \circ V : \mathcal{A} \to \mathcal{A}$. We already have the transformation $d : Y = U \circ V \to \mathrm{id}_{\mathcal{A}}$. Let us denote

$$s = 1_U \star \eta \star 1_V : Y = U \circ \mathrm{id}_{\mathcal{S}} \circ V \longrightarrow U \circ V \circ U \circ V = Y \circ Y$$

This places us in the situation of Example 0.2. It is immediate from the formulas that the X, d_i^n, s_i^n constructed above and the X, s_i^n, s_i^n constructed from Y, d, s in Example 0.2 agree. Thus, according to Lemma 0.3 it suffices to prove that

$$1_Y = (d \star 1_Y) \circ s = (1_Y \star d) \circ s \quad \text{and} \quad (s \star 1) \circ s = (1 \star s) \circ s$$

The first equal sign translates into the equality

$$1_U \star 1_V = (d \star 1_U \star 1_V) \circ (1_U \star \eta \star 1_V)$$

which holds if we have $1_U = (d \star 1_U) \circ (1_U \star \eta)$ which in turn holds by (0.1.1). Similarly for the second equal sign. For the last equation we need to prove

$$(1_U \star \eta \star 1_V \star 1_U \star 1_V) \circ (1_U \star \eta \star 1_V) = (1_U \star 1_V \star 1_U \star \eta \star 1_V) \circ (1_U \star \eta \star 1_V)$$

For this it suffices to prove $(\eta \star 1_V \star 1_U) \circ \eta = (1_V \star 1_U \star \eta) \circ \eta$ which is true because both sides are the same as $\eta \star \eta$. □